Newton School of Technology

Data Visualization & Analytics

Student Capstone Handbook

Execution + Expectations

1. Program Overview

Welcome to the **Sectoral Data Analytics Grand Capstone — Capstone 2**. This is not a standard college assignment; it is a 2-week industry simulation. You will operate as a professional analytics consultant team, taking a raw, messy dataset and converting it into actionable business intelligence — using **Python, GitHub, and Tableau** — under a tight deadline.

The 2-Week Sprint Model

Week 1: The Foundation (Engineering & Discovery)

- **Focus:** Data sourcing, Gate 1 approval, Python ETL pipeline, and initial EDA.
- **Milestone:** By the end of Week 1, your data must be clean, committed to GitHub, and analysis-ready.

Week 2: The Build (Analytics & Storytelling)

- **Focus:** Statistical analysis, KPI computation, Tableau dashboard, and final reporting.
- **Milestone:** Final submission — GitHub repo, Tableau URL, Project Report, and Presentation Deck.

2. Team Structure & Roles

Team Size: 6 to 7 Members

Formation: Teams are **formulated by the Faculty** to ensure balanced skill sets. No swapping is permitted.

Requirement: Every member must have visible GitHub commits and pull request contributions. Free-riding is detected via GitHub Insights and PR history.

While everyone is responsible for the final output, assign Leads to own specific phases to manage the compressed timeline:

Role	Primary Responsibility	Owner Name
Project Lead	Repo setup, timeline management, submission compliance, and ensuring the team hits Gate 1.	
Data Lead	Dataset sourcing, raw data commit to GitHub, and owning the data dictionary in docs/.	
ETL Lead	Notebooks 01 and 02 — Python extraction and cleaning pipeline.	
Analysis Lead	Notebooks 03 and 04 — EDA and statistical analysis in Python.	
Visualization Lead	Tableau dashboard design, publishing to Tableau Public, sharing URL and screenshots.	
Strategy Lead	Problem statement, KPI framework, and business recommendations.	
PPT & Quality Lead	Final report PDF, presentation deck, and contribution matrix.	

3. Execution Schedule (Day-by-Day)

Week 1: Data Sourcing, ETL & Cleaning

Objective: Secure mentor approval and prepare a clean, analysis-ready dataset committed to GitHub.

Day 1–2: Discovery & Setup

- **Task:** Meet your team, assign roles, select a sector (Retail, Finance, Health, etc.), and find a dataset.
- **Task:** Create the GitHub repository using the naming convention SectionName_TeamID_ProjectName and set up the folder structure.
- **Rule:** Dataset must be raw (min 5,000 rows, 8 columns). No pre-cleaned Kaggle competition files.

Day 3: Gate 1 — The "Go/No-Go" Checkpoint

- **Action:** Submit your Gate 1 proposal — problem statement, dataset link, and initial data dictionary.
- **Critical:** You cannot proceed to deep analysis until the mentor approves your dataset.
- **Backup:** Have at least two backup datasets ready from the same sector in case the primary is rejected.

Day 4–5: ETL Pipeline & Cleaning

- **Task:** Commit the raw dataset to data/raw/ — never edit this file.
- **Task:** Build the Python cleaning pipeline in notebooks/02_cleaning.ipynb. Log every transformation step.
- **Task:** Output cleaned data to data/processed/ and commit all notebooks to GitHub.
- **Deliverable:** A fully standardised dataset committed to GitHub, ready for analysis.

Week 2: Analysis, Dashboarding & Reporting

Objective: Uncover insights, build the Tableau dashboard, and deliver the final submission.

Day 6–7: EDA & Statistical Analysis

- **Task:** Run exploratory data analysis in notebooks/03_eda.ipynb — trends, distributions, outliers.
- **Task:** Perform statistical analysis in notebooks/04_statistical_analysis.ipynb — correlation, regression, hypothesis testing as applicable.
- **Task:** Compute KPIs and create final data load in notebooks/05_final_load_prep.ipynb.

Day 8–9: Tableau Dashboard

- **Task:** Build the final dashboard on Tableau Public. Design for decision-making, not decoration.
- **Requirement:** Must include interactive filters. Do not hard-code numbers.
- **Task:** Publish the dashboard, copy the public URL, and commit screenshots to tableau/screenshots/.

Day 10: Final Submission

- **Task:** Draft 3–5 specific, data-backed business recommendations.
- **Task:** Complete the final project report (PDF) and presentation deck.
- **Submission:** Ensure all assets are in the GitHub repo and all links work before the deadline.

4. Technical Requirements & Tooling

Tool	Status	Notes
Python + Jupyter	Mandatory	All ETL, cleaning, and statistical analysis must be in Python using Jupyter Notebooks (.ipynb files).
Google Colab	Supported	Teams may use Colab as their notebook environment. All .ipynb files must be committed to GitHub.
GitHub	Mandatory	All work must be version-controlled. Every member must have commits and PR contributions.
Tableau Public	Mandatory	All dashboards must be built and published on Tableau Public. Share the public URL and screenshots.
SQL	Optional	May be used for initial data extraction only. All SQL logic must be documented in the repo.

GitHub Repository Structure

All teams must maintain the following folder structure. Commit all notebooks, scripts, and reports here.

```
SectionName_TeamID_ProjectName/
- README.md
- data/
    - raw/        ← Original dataset (never edited)
    - processed/  ← Cleaned output from pipeline
- notebooks/
    - 01_extraction.ipynb
    - 02_cleaning.ipynb
    - 03_eda.ipynb
    - 04_statistical_analysis.ipynb
    - 05_final_load_prep.ipynb
- scripts/
    - etl_pipeline.py
- tableau/
    - screenshots/
    - dashboard_links.md
- reports/
    - project_report.pdf
    - presentation.pdf
- docs/
    - data_dictionary.md
```

GitHub Contribution Audit

Warning: The faculty will review GitHub Insights and PR history for every member. If a team member has no visible commits or pull requests, they may face an individual mark deduction.

5. Dataset Requirements

Minimum Standards

- **Row count:** At least 5,000 rows
- **Column count:** At least 8 meaningful analytical columns (numerical or categorical — not IDs or free text)
- **Quality:** Must contain realistic issues — missing values, inconsistent formats, or category variations that require cleaning
- **Format:** Tabular, row-level records suitable for Python ETL, KPI design, and Tableau visualisation

Approved Dataset Sources

- https://datasetsearch.research.google.com/
- https://www.kaggle.com/datasets (raw datasets only — see Kaggle rules below)
- https://data.gov.in/
- https://dataverse.harvard.edu/
- https://microdata.worldbank.org/
- https://archive.ics.uci.edu/
- https://data.world/ | https://datahub.io/

Special Rules for Kaggle Datasets

Kaggle contains both raw datasets and highly polished competition datasets. Most trending or competition-ready files are already clean and are not suitable for this capstone.

When using Kaggle, ensure:

- The dataset is raw or minimally processed
- Row-level records are available
- Missing values and inconsistencies are present
- No pre-built dashboard or feature-engineered version is used

Search using keywords such as: raw, original, transactional, logs, survey, records, events.

Note: Mentors may reject Kaggle datasets that are already analysis-ready.

Backup Dataset Requirement

Each team must submit one primary dataset and keep at least two backup options ready from the same sector. If the primary dataset is rejected at Gate 1, switch to a backup without delaying the timeline.

6. Submission Checklist

Before the final deadline, ensure your submission includes all of the following:

Asset 1: GitHub Repository

- ☐ Public repository created with correct naming convention
- ☐ All notebooks committed in .ipynb format
- ☐ data/raw/ contains the original, unedited dataset
- ☐ data/processed/ contains the cleaned output
- ☐ tableau/screenshots/ contains dashboard screenshots
- ☐ tableau/dashboard_links.md contains the Tableau Public URL
- ☐ docs/data_dictionary.md is complete
- ☐ README.md explains the project, dataset, and team
- ☐ All members have visible commits and pull requests

Asset 2: Tableau Dashboard

- ☐ Dashboard published on Tableau Public (accessible via public URL)
- ☐ At least one interactive filter included
- ☐ Dashboard addresses the business problem directly
- ☐ URL shared in dashboard_links.md and in the final report

Asset 3: Project Report (PDF) — 10 to 15 Pages

- ☐ Cover page with title, sector, team details, and institute
- ☐ Executive summary (1 page)
- ☐ Sector context and problem statement
- ☐ Data description — source, structure, size, and limitations
- ☐ Data cleaning and ETL methodology (Python pipeline)
- ☐ KPI and metric framework
- ☐ EDA with visualisations and written insights
- ☐ Statistical analysis results
- ☐ Dashboard design — screenshots and explanation
- ☐ 8–12 key insights written in decision language
- ☐ 3–5 actionable business recommendations
- ☐ Impact estimation, limitations, and future scope
- ☐ Contribution matrix (must match GitHub history)

Asset 4: Presentation Deck (PPT/PDF) — 10 to 12 Slides

- ☐ Slide 1: Title — project name, sector, team ID, members, mentor
- ☐ Slide 2: Context and problem statement
- ☐ Slide 3: Data engineering — source, size, cleaning steps, dictionary
- ☐ Slide 4: KPI and metrics framework
- ☐ Slide 5: Key insights from EDA (use business language)
- ☐ Slide 6: Advanced analysis — forecasting, segmentation, or root cause
- ☐ Slide 7: Dashboard walkthrough — executive and operational views
- ☐ Slide 8: Recommendations — 3–5 actionable, linked to insights

- ☐ Slide 9: Impact and value — cost, efficiency, revenue estimates
- ☐ Slide 10: Limitations and next steps

Asset 5: DVA-Oriented Resume & Portfolio (Individual)

- ☐ Resume updated to include this capstone project
- ☐ Portfolio link shared (if applicable)

7. Evaluation Rubric (100 Marks)

Area	Marks	Focus
Problem Framing	10	Is the business question clear and well-scoped?
Data Quality & ETL	15	Is the Python pipeline thorough and documented?
Analysis Depth	25	Are statistical methods applied correctly with insight?
Dashboard & Visualization	20	Is the Tableau dashboard interactive and decision-relevant?
Business Recommendations	20	Are insights actionable and well-reasoned?
Storytelling & Clarity	10	Is the presentation professional and coherent?

Remember: Marks are awarded for analytical thinking and decision relevance — not chart quantity, visual decoration, or code length.

8. What Makes a High-Scoring Project

- A clear, well-scoped business problem statement
- A documented, reproducible Python ETL pipeline in Jupyter Notebooks
- Strong statistical analysis — not just counts and averages
- A Tableau dashboard that answers real business questions
- Insights written in decision language, not chart descriptions
- Practical, specific recommendations with estimated business impact
- A professional, coherent presentation deck and final report
- Every team member with visible GitHub contributions

The difference between an Observation and an Insight:

Observation (Not enough)	Insight (What we want)
"Sales dropped in May."	"Sales dropped in May due to stockouts, requiring a change in re-order points."

"Customer churn is high in Q3."	"Q3 churn correlates with delayed deliveries — a logistics SLA fix could recover 12% of lost customers."

9. Rules & Academic Integrity

- No plagiarism — all analysis, code, and recommendations must be your team's original work
- No copying from other teams — projects are compared during evaluation
- Every team member must understand the dataset, cleaning steps, KPIs, and dashboard structure
- Viva questions can be directed to any team member — not just the presenter
- Late submission will incur a grade penalty
- Copied dashboards, reused templates without understanding, or duplicate work across teams will be penalised

Contribution Matrix: A contribution matrix must be included in the final report and must match the evidence in GitHub Insights, PR history, and committed files.

10. How This Helps Your Career

- This becomes your primary interview project — a real end-to-end analytics case
- Demonstrates Python-based ETL, statistical analysis, and Tableau skills in one portfolio piece
- Your GitHub repository is a live portfolio — keep it clean and well-documented
- The Tableau Public dashboard is publicly accessible — share the link on LinkedIn and your resume
- Shows real business thinking — the ability to convert data into decisions
- Improves your confidence in explaining data, methodology, and recommendations under pressure

Tools don't get you marks. Thinking gets you marks.
Dashboards don't get you marks. Decisions get you marks.